

MUHAMMAD ABDULLAH ANWAR

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RESEARCH PROFILE

Master's student in Energy Systems specializing in smart grids, secure and resilient power system operation, renewable integration, optimization, control, and AI/ML applications for modern energy systems. Research experience spans cyber-physical security, game-theoretic modeling of false data injection attacks, optimal power flow, grid-forming inverter control, adaptive droop control, relay protection testing, and graph neural network-based OPF. Strong interest in academic research internships, PhD-level research, and engineering roles at the intersection of power systems, power electronics, optimization, and intelligent energy infrastructure.

EDUCATION

Skolkovo Institute of Science and Technology, Skoltech

Moscow, Russia

Master of Science in Energy Systems; Current CGPA: 3.7/4.0

Sep. 2024 – Jun. 2026

- Research focus: smart grids, optimization, power systems, power electronics, power system dynamics and stability, renewable energy integration, optimal control, and cybersecurity in smart grids.

Comsats University Islamabad

Lahore, Pakistan

Bachelor of Science in Electrical Engineering

Sep. 2019 – Jul. 2023

- Academic focus: power electronics, IoT, signal processing, complex electrical circuit analysis, control systems, communication systems, and digital system design.

PUBLICATIONS AND MANUSCRIPTS

An AC-OPF-Based Stackelberg Framework for Cyber-Physical Attack Analysis in Power Systems

Manuscript submitted to Elsevier

Submitted

- Developed an AC-OPF-based Stackelberg optimization framework for cyber-physical attack analysis in power systems.

An Optimal Contingency-Sensitive Inertia and Damping Control for Grid-Forming Inverters

7th International Conference on Control Systems, Mathematical Modeling, Automation and Energy Efficiency, SUMMA 2025

- Accepted for presentation/publication at SUMMA 2025, Lipetsk, Russia. Available on IEEE Xplore.

Optimal Adaptive Droop Control with AI-Based Contingency Detection for Frequency Regulation

Manuscript submitted to IEEE Transactions on Power Systems

Submitted

- Developed from research on adaptive droop control, optimization-based stability constraints, and AI-assisted contingency detection for frequency regulation.

RESEARCH EXPERIENCE

Master's Thesis: A Game Theory-based Optimization Framework for Cybersecurity in Power Systems under False Data Injection Attacks

Skoltech, Moscow, Russia

Sep. 2025 – Present

- Developed Stackelberg and Nash game-theoretic models to analyze attacker-defender interactions under false data injection attacks in power systems.
- Formulated an AC-OPF-based cyber-physical attack analysis framework to evaluate adversarial impacts on secure power system operation.
- Identified critical buses and designed defense optimization strategies to reduce attack impact and strengthen cyber-physical resilience.

Security-Constrained Multi-Period Optimal Power Flow with Renewable and Storage Integration

Skoltech, Moscow, Russia

Nov. 2025 – Jan. 2026

- Developed a multi-period OPF model incorporating renewable generation, energy storage, ramping limits, and unit commitment constraints.
- Implemented the optimization framework in Pyomo/IPOPT and evaluated generation cost, locational marginal prices, and computational performance on IEEE test systems.

Graph Neural Network Approach for Optimal Power Flow

Skoltech, Moscow, Russia

Sep. 2025 – Nov. 2025

- Built a graph neural network surrogate model for AC optimal power flow on the IEEE 9-bus system using PyTorch Geometric.
- Generated OPF datasets under varying load conditions and evaluated prediction accuracy for bus voltages and generator dispatch.

Contingency-Sensitive Inertia and Damping Control for Grid-Forming Inverters

Skoltech, Moscow, Russia

Jun. 2025 – Sep. 2025

- Designed a contingency-sensitive control strategy to adapt grid-forming inverter inertia and damping for improved transient stability.
- Implemented an offline-online optimization framework and validated the method on the IEEE 9-bus system using RTDS NovaCor.

Optimal Adaptive Droop Control with AI-Based Contingency Detection for Frequency Regulation

Skoltech, Moscow, Russia

Jan. 2025 – May 2025

- Developed an adaptive droop control framework for primary frequency regulation under changing operating conditions.
- Integrated optimization, eigenvalue-based stability constraints, and AI-based event detection to coordinate system response.

Bachelor Thesis: IoT-Based Smart Energy Metering and Monitoring System

Comsats University Islamabad, Lahore, Pakistan

Sep. 2022 – Jul. 2023

- Designed an IoT-based smart energy meter with real-time data acquisition, cloud transmission, and a web dashboard for energy monitoring.

TECHNICAL SKILLS

Programming: Python, MATLAB, C++, C, Java, Julia, R
AI/ML and Data Analysis: PyTorch, TensorFlow, PyTorch Geometric, NumPy, Pandas, OpenCV, Matplotlib
Optimization and Power Systems: Pyomo, IPOPT, MATPOWER, ETAP, PSCAD, Simulink, optimal power flow, security-constrained OPF, inverter control
Tools and Environments: Git/GitHub, Jupyter, Wireshark, Linux, LaTeX, RTDS NovaCor
Signal Processing: Sampling, quantization, digital filtering, spectral denoising, transform-based compression

WORK EXPERIENCE

Research Engineer Jun. 2025 – Present
Tekvel Moscow, Russia – Hybrid
• Develop and implement relay protection test plans to safeguard electrical networks, improving system reliability and fault response.
• Conduct fault simulations and coordinate with protection and control teams to evaluate relay behavior and refine protection settings.
• Ensure compliance with engineering standards by documenting test results, identifying technical risks, and recommending system improvements.

Design Engineer Nov. 2023 – Sep. 2024
Telerelations Lahore, Pakistan
• Delivered controls design and technical documentation for a New York City-based client, aligning outputs with NYC codes/standards and project requirements.
• Designed SCA, DDC, and DASSY controls packages for Building Automation/BMS projects, supporting control execution and installation workflows.

CONFERENCES AND SEMINARS

Research Seminars in Smart Grids Nov. 2024 – May 2025
Skolkovo Institute of Science and Technology, Moscow, Russia
• Engaged in research seminars on smart grid control, optimization, energy system stability, and advanced methods for modern power systems.

7th International Conference on Control Systems, Mathematical Modeling, Automation and Energy Efficiency, SUMMA 2025 Oct. 2025
Conference, Lipetsk, Russia

AWARDS AND SCHOLARSHIPS

Skoltech Merit Scholarship 2024 – Present
Skolkovo Institute of Science and Technology, Skoltech Moscow, Russia
• Awarded a full scholarship for outstanding academic record and research excellence in Electrical Engineering.

Excellence Award for Outstanding Industrial Immersion Project Oct. 2025
Skolkovo Institute of Science and Technology, Skoltech Moscow, Russia
• Recognized for delivering an innovative and high-impact Industrial Immersion project.

LANGUAGES

English: C1 **Urdu:** Native

ACADEMIC REFERENCES

Prof. Oleg Khamisov
Research Advisor, Master in Energy Systems
Skolkovo Institute of Science and Technology, Moscow, Russia
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Dr. Naeem Shahzad
Bachelor Supervisor, Department of Electrical Engineering
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